

STATS 7022 - Data Science

Bird Bath Analysis

Assignment 2: Question 1

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1 Load Libraries

2 Load Data

The data is contained within the file `bird_bath.rds`. The data is read in to `R`.

There are 161,057 rows in the `bird_bath` data.

3 Data Cleaning

3.1 Remove Missing Years

Some rows are missing a survey year (`survey_year`). These rows are removed from the data.

There are now 160,888 rows in the `bird_bath` data.

3.2 Identify the Top 15 Types of Bird

We consider only the top 15 bird species (`bird_type`) by frequency (`bird_count`). Results are summarised in Table 3.1.

Table 3.1: Top 15 birds at Australian bird baths, by frequency.

bird_type	count
Noisy Miner	292
Australian Magpie	262
Rainbow Lorikeet	235
Red Wattlebird	171
Superb Fairy-wren	170
Magpie-lark	160
Pied Currawong	157
Crimson Rosella	155
Eastern Spinebill	153
Spotted Dove	151
Lewin's Honeyeater	133

bird_type	count
Satin Bowerbird	131
Crested Pigeon	125
Grey Fantail	123
Red-browed Finch	117

There are now 14,280 rows in the `bird_bath` data.

3.3 Create the Response Variable

We define a new response variable, called `bird_present`, using the following rule:

- If `bird_count` is greater than zero for a row, then `bird_present` takes value “yes”,
- Otherwise, `bird_present` takes value “no”.

3.4 Select Columns

We only need the following columns:

- `urban_rural`
- `bird_type`
- `bird_present`

All other columns are removed.

3.5 Convert Variables to Categorical

We convert all remaining variables to categorical variables.

4 Models

We create two separate workflows using `tidymodels`:

1. A logistic regression model with `bird_present` as the response variable and `urban_rural` and `bird_type` as the predictors.

```
## —— Workflow —————
##
## Preprocessor: Recipe
## Model: logistic_reg()
##
## —— Preprocessor —————
##
## 1 Recipe Step
##
## • step_dummy()
##
## —— Model —————
##
## Logistic Regression Model Specification (classification)
##
## Computational engine: glm
```

2. A logistic regression model with the same predictors and response, plus a two-way interaction between `urban_rural` and `bird_type`.

```
## --- Workflow -----  
##  
## Preprocessor: Recipe  
## Model: logistic_reg()  
##  
## --- Preprocessor -----  
##  
## 2 Recipe Steps  
##  
## • step_dummy()  
## • step_interact()  
##  
## --- Model -----  
##  
## Logistic Regression Model Specification (classification)  
##  
## Computational engine: glm
```

Both models are fitted to the `bird_bath` data, with ROC curves displayed in Figure 5.1.

5 ROC Curves

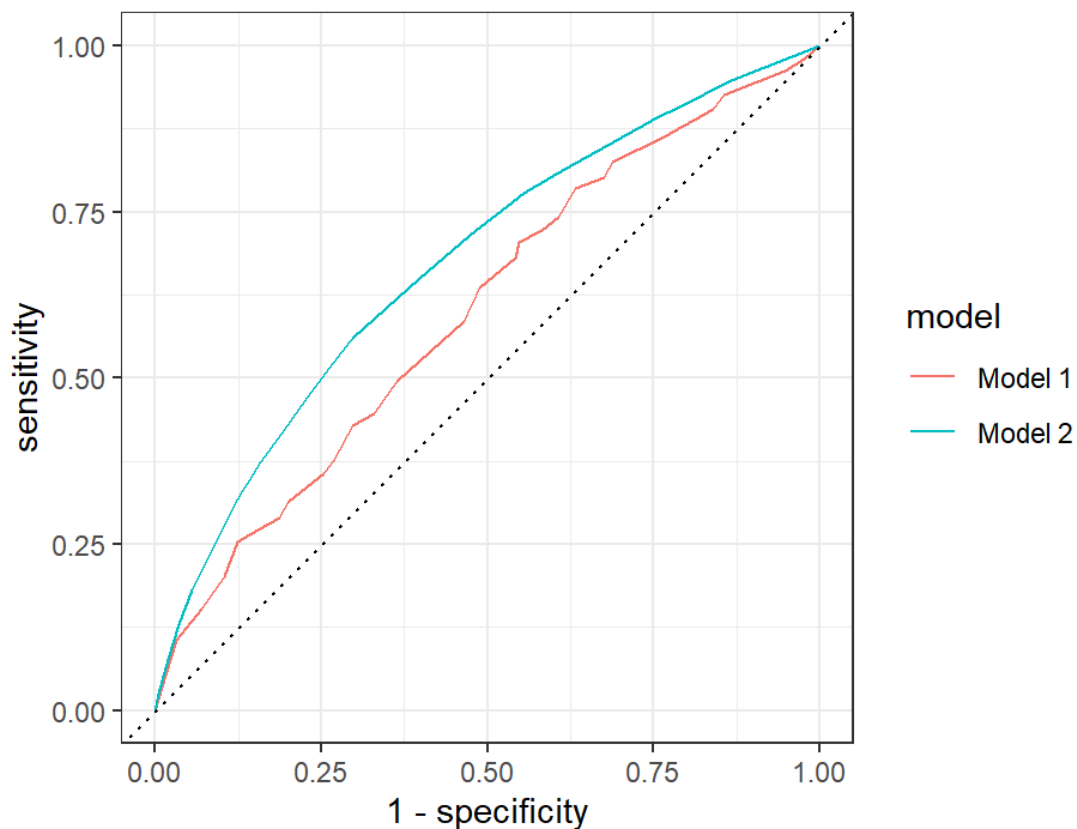


Figure 5.1: ROC curves for logistic regression models.